

CHAPTER 13

Analysis of Covariance

Predictors that are qualitative in nature, for example, eye color, are sometimes described as *categorical* or called *factors*. We wish to incorporate these predictors into the regression analysis. Analysis of covariance (ANCOVA) refers to regression problems where there is a mixture of quantitative and qualitative predictors.

Suppose we are interested in the effect of a medication on cholesterol level. We might have two groups; one receives the medication and the other, the default treatment. We could not treat this as a simple two-sample problem if we knew that the two groups differed with respect to age, as this would affect the cholesterol level. See Figure 13.1 for a simulated example. For the patients who received the medication, the mean reduction in cholesterol level was 0% while for those who did not, the mean reduction was 10%. So superficially it would seem that it would be better not to be treated. However, the treated group ranged in age from 50 to 70 while those who were not treated ranged in age between 30 and 50. We can see that once age is taken into account, the difference between treatment and control is again 10%, but this time in favor of the treatment.

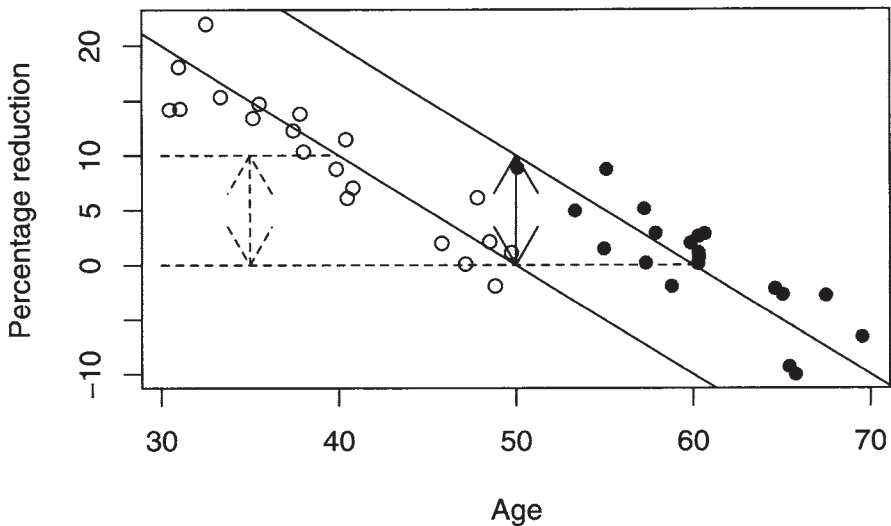


Figure 13.1 *Simulated example showing the confounding effect of a covariate. The patients who took the medication are marked with a solid dot while those who did not are marked with an empty dot.*

ANCOVA adjusts the groups for the age difference and then presents the effect of the medication. It can also be used when there are more than two groups and more than one covariate.

Our strategy is to incorporate the qualitative predictors within the $y = X\beta + \epsilon$ framework. We can then use the estimation, inferential and diagnostic techniques that we have already learned. This avoids having to learn a different set of formulae for each new type of qualitative predictor configuration. To put qualitative predictors into the $y = X\beta + \epsilon$ form

we need to code the qualitative predictors. We consider a specific example where 3; is the change in cholesterol level, x is the age and:

$$d = \begin{cases} 0 & \text{did not take medication} \\ 1 & \text{took medication} \end{cases}$$

d is called a *dummy variable*. A variety of linear models may be considered here:

1. The same regression line for both groups: $y = \beta_0 + \beta_1 x + \epsilon$ or is written in R as $y \sim x$.
2. Separate regression lines for each group with the same slope: $y = \beta_0 + \beta_1 x + \beta_2 d + \epsilon$ or is written in R as $y \sim x + d$. In this case β_2 represents the vertical distance between the regression lines (i.e., the effect of the drug).
3. Separate regression lines for each group with the different slopes: $y = \beta_0 + \beta_1 + \beta_2 d + \beta_3 x \cdot d + \epsilon$ or is written in R as $y \sim x + d \mid x$ or $y \sim x * d$. To form the slope interaction term $d : x$ in the X -matrix, multiply x by d elementwise. Any interpretation of the effect of the drug will now also depend on age.

Estimation and testing work just as they did before. Interpretation is much easier if we can eliminate the slope interaction term.

Other codings of d are possible, for instance:

$$d = \begin{cases} -1 & \text{did not take medication} \\ 1 & \text{took medication} \end{cases}$$

is used by some. This coding enables β_2 and β_3 to be viewed as differences from a response averaged over the two groups. Any other coding that assigned a different number to the two groups would also work, but interpretation of the estimated parameters would be more difficult.

13.1 A Two-Level Example

The data for this example come from a study of the effects of childhood sexual abuse on adult females reported in Rodriguez et al. (1997): 45 women treated at a clinic, who reported childhood sexual abuse (csa), were measured for post-traumatic stress disorder (ptsd) and childhood physical abuse (cpa) both on standardized scales. 31 women treated at the same clinic, who did not report childhood sexual abuse, were also measured. The full study was more complex than reported here and so readers interested in the subject matter should refer to the original article.

We take a look at the data and produce a summary subsetted by csa:

```
> data (sexab)
> sexab
      cpa      ptsd      csa
1  2.04786  9.71365 Abused
```

```

2 0.83895 6.16933 Abused
....
75 2.85253 6.84304 NotAbused
76 0.81138 7.12918 NotAbused
> by (sexab, sexab$csa, summary)
sexab$csa: Abused
      Cpa      ptsd      csa
Min.   : -1.11 Min.   : 5.98 Abused   :45
1st Qu.: 1.41 1st Qu.: 9.37 NotAbused: 0
Median : 2.63 Median : 11.31
Mean    : 3.08 Mean    : 11.94
3rd Qu.: 4.32 3rd Qu.: 14.90
Max.    : 8.65 Max.    : 18.99
sexab$csa: NotAbused
      Cpa      ptsd      csa
Min.   : -3.12 Min.   : -3.35 Abused   : 0
1st Qu.: -0.23 1st Qu.: 3.54 NotAbused:31
Median : 1.32 Median : 5.79
Mean    : 1.31 Mean    : 4.70
3rd Qu.: 2.83 3rd Qu.: 6.84
Max.    : 5.05 Max.    : 10.91

```

Now plot the data—see Figure 13.2:

```

> plot (ptsd ~ csa, sexab)
> plot (ptsd ~ cpa, pch=as.character (csa), sexab)

```

We see that those in the abused group have higher levels of PTSD than those in the nonabused in the left panel of Figure 13.2. We can test this difference:

```

> t.test(sexab$ptsd[1:45],sexab$ptsd[46:76])
      Welch Two Sample t-test
data:  sexab$ptsd[1:45] and sexab$ptsd[46:76]
t = 8.9006, df = 63.675, p-value = 8.803e-13
alt. hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 5.6189 8.8716
sample estimates:
mean of x mean of y
 11.9411    4.6959

```

and find that it is clearly significant. However, in the right panel of Figure 13.2 we see that there is positive correlation between PTSD and childhood *physical* abuse and in the numerical summary we see that those in the abused group suffered higher levels (3.08 vs. 1.31) of cpa than those in the nonabused group. This suggests physical abuse as an alternative explanation of higher PTSD in the sexually abused group.

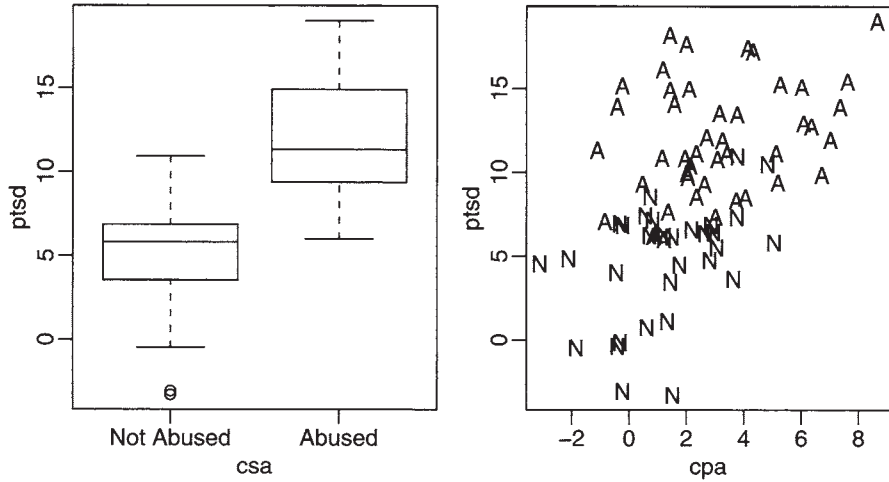


Figure 13.2 PTSD comparison of abused and nonabused subjects on the left. A=Abused and N=NotAbused on the right.

ANCOVA allows us to disentangle these two competing explanations. We fit the separate regression lines model. `ptsd ~ cpa*csa` is an equivalent model formula:

```
> g <- lm (ptsd cpa+csa+cpa:csa, sexab)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    10.557      0.806   13.09 < 2e-16
cpa              0.450      0.208    2.16  0.034
csaNotAbused   -6.861      1.075   -6.38 1.5e-08
cpa:csaNotAbused 0.314      0.368    0.85 0.397
Residual standard error: 3.28 on 72 degrees of freedom
Multiple R-Squared: 0.583, Adjusted R-squared: 0.565
F-statistic: 33.5 on 3 and 72 DF, p-value: 1.13e-13
```

Because `csa` is nonnumeric, R automatically treats it as a qualitative variable and sets up a coding. We can discover the coding by examining the *X*-matrix:

```
> model.matrix (g)
(Intercept)      cpa csaNotAbused cpa: csaNotAbused
1           1 2.04786              0           0.00000
2           1 0.83895              0           0.00000
.....
75          1 2.85253              1           2.85253
76          1 0.81138              1           0.81138
```

We see that “Abused” is coded as zero and “NotAbused” is coded as one. The default choice is made alphabetically. This means that “Abused” is the *reference level* here and that the parameters represent the difference between “NotAbused” and this reference level. In this case, it would be slightly more convenient if the coding was reversed. The interaction term `cpa: csaNotAbused` is represented in the fourth column of the matrix as

the product of the second and third columns which represents the terms from which the interaction is formed.

We see that the model can be simplified because the interaction term is not significant. We reduce to this model:

```
> g <- lm (ptsd ~ cpa+csa,sexab)
> summary (g)
Coefficients:
              Estimate   Std. Error t value Pr(>|t|)
(Intercept)   10.248         0.719   14.26  <2e-16
cpa             0.551         0.172    3.21   0.002
csaNotAbused  -6.273         0.822   -7.63  6.9 e-11
Residual standard error: 3.27 on 73 degrees of freedom
Multiple R-Squared: 0.579, Adjusted R-squared: 0.567
F-statistic: 50.1 on 2 and 73 DF, p-value: 2e-14
```

No further simplification is possible because the remaining predictors are statistically significant.

Put the two parallel regression lines on the plot, as seen in the left panel of Figure 13.3.

```
> plot (ptsd ~ cpa, pch=as.character (csa), sexab)
> abline (10.248, 0.551)
> abline (10.248-6.273, 0.551, lty=2)
```

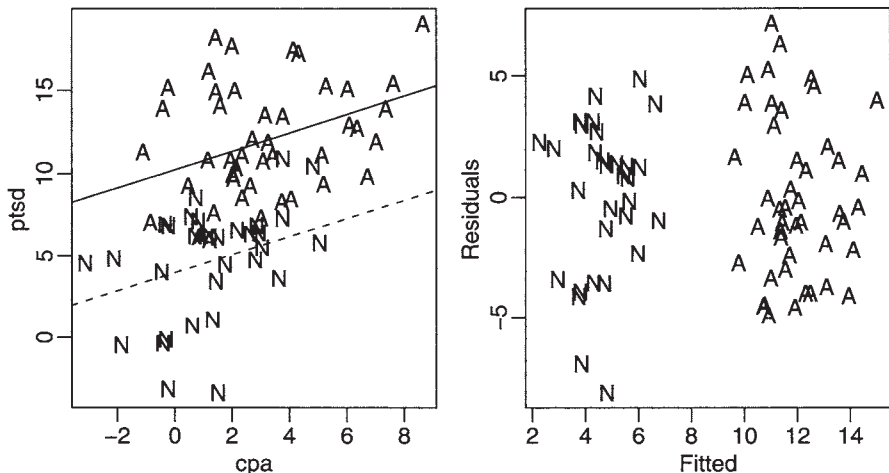


Figure 13.3 Model fit shown on the left and fitted vs. residuals plot on the right. A=Abused and N=Not Abused.

The slope of both lines is 0.551, but the “Abused” line is 6.273 higher than the “NonAbused.” From the t-test above, the estimated effect of childhood sexual abuse is $11.9411 - 4.6959 = 7.2452$. So after adjusting for the effect of childhood physical abuse, our estimate of the effect of childhood sexual abuse on PTSD is mildly reduced.

We can also compare confidence intervals for the effect of csa:

```
> confint (g) [3,]
      2.5 %      97.5 %
-7.9108    -4.6347
```

compared to the (5.6189, 8.8716) found for the unadjusted difference. In this particular case, the confidence intervals are about the same width. In other cases, particularly designed experiments, adjusting for a covariate can increase the precision of the estimate of an effect.

The usual diagnostics should be checked. It is worth checking whether there is some difference related to the categorical variable as we do here:

```
> plot      (fitted      (g),      residuals      (g),      pch=as.character
      (sexab$csa), xlab="Fitted", ylab="Residuals")
```

We see in the right panel of Figure 13.3 that there are no signs of heteroscedasticity. Furthermore, because the two groups happen to separate, we can also see that the variation in the two groups is about the same. If this were not so, we would need to make some adjustments to the analysis, possibly using weights.

For convenience, you can change the reference level:

```
> sexab$csa <- relevel (sexab$csa, ref="NotAbused")
> g <- lm (ptsd ~ cpa+csa,sexab)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      3.975      0.629      6.32 1.9e-08
cpa              0.551      0.172      3.21 0.002
csaAbused        6.273      0.822      7.63 6.9e-11
Residual standard error: 3.27 on 73 degrees of freedom
Multiple R-Squared: 0.579, Adjusted R-squared: 0.567
F-statistic: 50.1 on 2 and 73 DF, p-value: 2e-14
```

Although some of the coefficients have different numerical values, this coding leads to the same conclusion as before.

Finally, we should point out that childhood physical abuse might not be the only factor that is relevant to assessing the effects of childhood sexual abuse. It is quite possible that the two groups differ according to other variables such as socioeconomic status and age. Issues such as these were addressed in Rodriguez et al. (1997).

13.2 Coding Qualitative Predictors

A more extensive use of dummy variables is needed for factors with more than two levels. Let B be an $n \times k$ dummy variable matrix where $B_{ij}=1$ if case i falls in class j and is zero otherwise. We might use B to form part of the model matrix. However, the row sums of B are all one. Since an intercept term would also be represented by a column of ones, all the parameters would not be identifiable.

Removing the intercept term is one solution, but this will not work well if there is more than one factor. A more general solution is to reduce the rank of the dummy variable matrix. Simply deleting one column would do, but any solution that avoids collinear dummy variables will work. The choice should be based on convenience of interpretation and numerical stability.

The coding is determined by a *contrast matrix* C which has dimension $k \times (k-1)$. Some examples of C are given below. The contribution to the model matrix is then given by BC . Other columns of the model matrix might include a column of ones for the intercept and perhaps other predictors.

Treatment coding

Consider a four-level factor that will be coded using three dummy variables. This contrast matrix describes the coding, where the columns represent the dummy variables and the rows represent the levels:

```
> contr.treatment (4)
  2 3 4
1 0 0 0
2 1 0 0
3 0 1 0
4 0 0 1
```

This treats level one as the standard level to which all other levels are compared so a control group, if one exists, would be appropriate for this level. The parameter for the dummy variable then represents the difference between the given level and the first level. R assigns levels to a factor in alphabetical order by default. Treatment coding is the default choice for R.

Helmert coding

The coding here is:

```
> contr.helmert (4)
  [,1] [,2] [,3]
1  -1  -1  -1
2   1  -1  -1
3   0   2  -1
4   0   0   3
```

If there are equal numbers of observations in each level (a balanced design) then the dummy variables will be orthogonal to each other and to the intercept. This coding is not as nice for interpretation except in some special cases. It is the default choice in S-plus.

There are other choices of coding—anything that spans the $k-1$ dimensional space will work. The choice of coding does not affect the R^2 , $\hat{\sigma}^2$ and overall F -statistic. It does affect the $\hat{\beta}$ and you do need to know what the coding is before making conclusions about $\hat{\beta}$.

When there is an interaction between two terms represented by model matrix components X_1 and X_2 , we compute the element-wise product of every column of X_1 with every column of X_2 to get the representation of the interaction in the model matrix.

13.3 A Multilevel Factor Example

The data for this example come from a study on the sexual activity and the life span of male fruitflies by Partridge and Farquhar (1981): 125 fruitflies were divided randomly into five groups of 25 each. The response was the longevity of the fruitfly in days. One group was kept solitary, while another was kept individually with a virgin female each day. Another group was given eight virgin females per day. As an additional control, the fourth and fifth groups were kept with one or eight pregnant females per day. Pregnant fruitflies will not mate. The thorax length of each male was measured as this was known to affect longevity. The five groups are labeled many, isolated, one, low and high respectively. The purpose of the analysis is to determine the difference between the five groups if any. We start with a plot of the data, as seen in Figure 13.4.

```
> data (fruitfly)
> plot (longevity ~ thorax, fruitfly, pch=unclass (activity))
> legend (0.63, 100, levels (fruitfly$activity), pch=1:5)
```

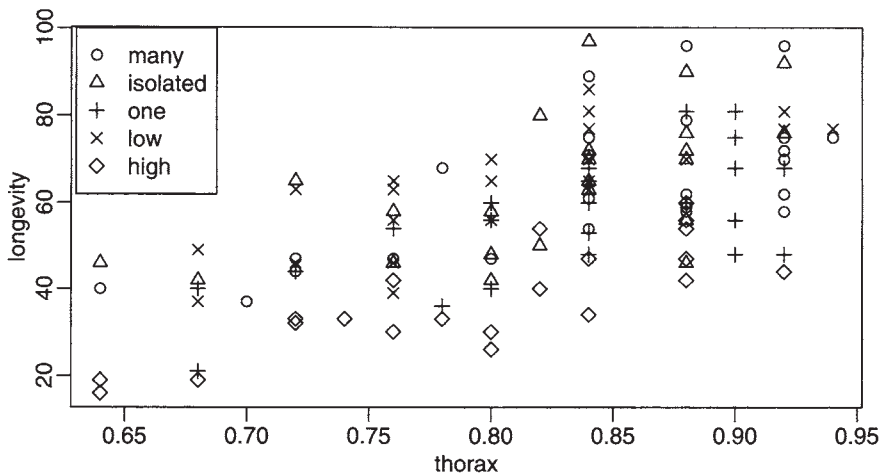


Figure 13.4 *Plot of longevity in days and thorax length in mm of fruitflies divided into five treatment groups. Longevity for the high sexual activity group appears to be lower.*

We fit and summarize the most general linear model:

```
> g <- lm (longevity ~ thorax*activity, fruitfly)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -50.242     21.801   -2.30    0.023
thorax          136.127     25.952    5.25 7.3e-07
activityisolated   6.517     33.871    0.19  0.848
activityone      -7.750     33.969   -0.23  0.820
activitylow      -1.139     32.530   -0.04  0.972
```

```

activityhigh          -11.038    31.287    -0.35    0.725
thorax:activityisolated -4.677    40.652    -0.12    0.909
thorax:activityone      0.874    40.425     0.02    0.983
thorax:activitylow      6.548    39.360     0.17    0.868
thorax:activityhigh    -11.127    38.120    -0.29    0.771
Residual standard error: 10.7 on 114 degrees of freedom
Multiple R-Squared: 0.653, Adjusted R-squared: 0.626
F-statistic: 23.9 on 9 and 114 DF, p-value: < 2e-16

```

Since “many” is the reference level, the fitted regression line within this group is $\text{longevity} = -50.2 + 136.1 \cdot \text{thorax}$. For “isolated,” it is $\text{longevity} = (-50.2 + 6.5) + (136.1 - 4.7) \cdot \text{thorax}$. Similar calculations can be made for the other groups. Examine:

```
> model.matrix (g)
```

to see how the coding is done. Some diagnostics should be examined by:

```
> plot (g)
```

There is perhaps some mild heteroscedasticity, but we will let this be until later for ease of presentation. Now see whether the model can be simplified:

```

> anova (g)
Analysis of Variance Table Response: longevity
      Df Sum Sq Mean Sq F value Pr(>F)
thorax    1  15003   15003   130.73 < 2e-16
activity   4   9635    2409    20.99 5.5e-13
thorax:activity 4     24      6     0.05      1
Residuals 114  13083    115

```

This is a sequential analysis of variance (ANOVA) table. Starting from a null model, terms are added and sequentially tested. The models representing the null and alternatives are listed in Table 13.1. In ANCOVA, we wish to successively simplify the full model

Null	Alternative
$y \sim 1$	$y \sim \text{thorax}$
$y \sim \text{thorax}$	$y \sim \text{thorax} + \text{activity}$
$y \sim \text{thorax} + \text{activity}$	$y \sim \text{thorax} + \text{activity} + \text{thorax:activity}$

Table 13.1 *Models compared in the sequential ANOVA.*

and then interpret the result. The interaction term `thorax: activity` is not significant indicating that we can fit the same slope within each group. No further simplification is possible.

We notice that the F-statistic for the test of the interaction term is very small and so the p-value is not exactly one (due to the rounding), but is very close:

```
> 1-pf (0.05, 4, 114)
[1] 0.99525
```

For these data, the fitted regression lines to the five groups happen to be very close to parallel. This can, of course, just happen by chance. In some other cases, unusually large p-values have been used as evidence that data have been tampered with or “cleaned” to improve the fit. Most famously, Ronald Fisher suspected Gregor Mendel of fixing the data in some genetics experiments because the data seemed too good to be true. See Fisher (1936).

We now refit without the interaction term:

```
> gb <- lm (longevity ~ thorax+activity, fruitfly)
```

Do we need both thorax and activity? We could use the output above which suggests both terms are significant. However, thorax is tested by itself and then activity is tested once thorax is entered into the model. We might prefer to check whether each predictor is significant once the other has been taken into account. We can do this using:

```
> drop1 (gb, test="F")
Single term deletions
Model:
longevity ~ thorax + activity
      Df Sum of Sq  RSS AIC F value    Pr (F)
<none>                 13107 590
thorax    1      12368 25476 670    111.3 < 2e-16
activity  4       9635 22742 650     21.7 2.0e-13
```

The `drop1 ( )` command tests each term relative to the full model. This shows that both terms are significant even after allowing for the effect of the other. Now examine the model coefficients:

```
> summary (gb)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -48.75      10.85   -4.49  1.6e-05
thorax         134.34      12.73   10.55 < 2e-16
activityisolated 2.64       2.98    0.88  0.379
activityone    -7.01       2.98   -2.35  0.020
activitylow     4.14       3.03    1.37  0.174
activityhigh   -20.00       3.02   -6.63  1.0e-09
Residual standard error: 10.5 on 118 degrees of freedom
Multiple R-Squared:  0.653, Adjusted R-squared:  0.638
F-statistic: 44.4 on 5 and 118 DF, p-value: < 2e-16
```

“Many” is the reference level. We see that the intercepts of “isolated” and “low” are not significantly different from many. We also see that the low sexual activity group, “one,” survives about seven days less. The p-value is 0.02 and is enough for statistical significance if only one comparison is made. However, we are making more than one comparison, and so, as with outliers, a Bonferroni-type adjustment might be considered. This would erase the statistical significance of the difference.

However, the high sexual activity group, "many," has a life span 20 days less than the reference group and this is strongly significant.

Returning to the diagnostics:

```
> plot (residuals (gb) ~ fitted (gb),
       pch=unclass (fruitfly$activity))
```

is seen in the first panel of Figure 13.5. We have some nonconstant variance although it does not appear to be related to the five groups. A log transformation can remove the heteroscedasticity:

```
> gt <- lm (log (longevity) ~ thorax+activity, fruitfly)
> plot (residuals (gt) ~ fitted (gt),
       pch=unclass (fruitfly$activity))
```

as seen in the second panel of Figure 13.5. One disadvantage of transformation is

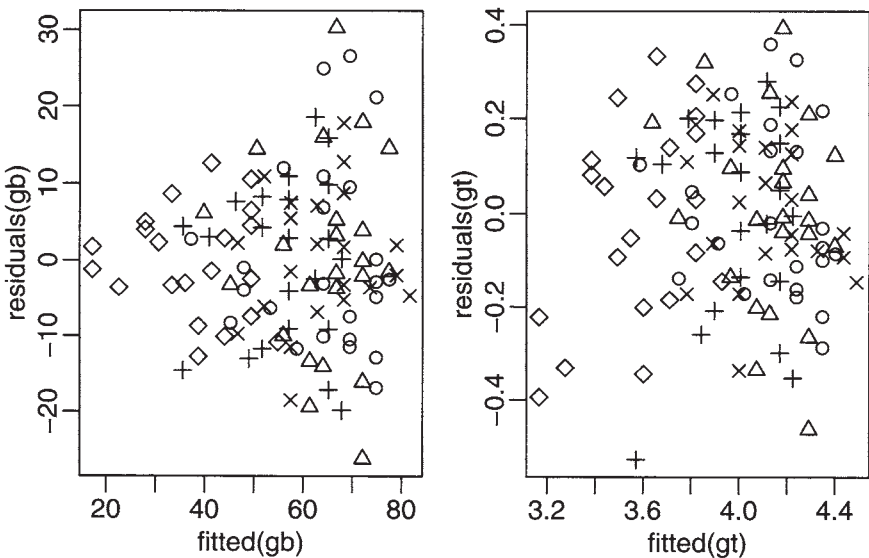


Figure 13.5 *Diagnostic plots for the fruitfly data before and after log transformation of the response.*

that it can make interpretation of the model more difficult. Let's examine the model fit:

```
> summary (gt)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.8442     0.1988   9.28 1.0e-1
thorax          2.7215     0.2333  11.67 < 2e-1
activityisolated 0.0517     0.0547   0.95  0.34
activityone     -0.1239     0.0546  -2.27  0.02
activitylow      0.0879     0.0555   1.59  0.116
```

```

activityhigh      -0.4193      0.0553      -7.59      8.4e-1
Residual standard error: 0.193 on 118 degrees of freedom
Multiple R-Squared: 0.702, Adjusted R-squared: 0.69
F-statistic: 55.7 on 5 and 118 DF, p-value: < 2e-16

```

Notice that the R^2 is higher for this model, but the p-values are similar. Because of the log transformation, we can interpret the coefficients as having a multiplicative effect:

```

> exp (coef (gt) [3:6])
activityisolated activityone activitylow activityhigh
      1.05311      0.88350      1.09189      0.65754

```

Compared to the reference level, we see that the high sexual activity group has 0.66 times the life span (i.e, 34% less).

Why did we include thorax in the model? Its effect on longevity was known, but because of the random assignment of the flies to the groups, this variable will not bias the estimates of the effects of the activities. We can verify that thorax is unrelated to the activities:

```

> gh <- lm (thorax ~ activity, fruitfly)
> anova (gh)
Analysis of Variance Table
Response: thorax
      Df      Sum Sq      Mean Sq    F value    Pr(>F)
activity    4       0.026       0.006       1.11     0.36
Residuals 119       0.685       0.006

```

However, look what happens if we omit thorax from the model for longevity:

```

> gu <- lm (log (longevity) ~ activity, fruitfly)
> summary (gu)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.1193    0.0564   72.99 < 2e-16
activityisolated 0.0234    0.0798    0.29   0.77
activityone    -0.1195    0.0798   -1.50   0.14
activitylow     0.0240    0.0806    0.30   0.77
activityhigh   -0.5172    0.0798   -6.48 2.2e-09
Residual standard error: 0.282 on 119 degrees of freedom
Multiple R-Squared: 0.359, Adjusted R-squared: 0.338
F-statistic: 16.7 on 4 and 119 DF, p-value: 6.96e-11

```

The magnitude of the effects do not change that much but the standard errors are substantially larger. The value of including thorax in this model is to increase the precision of the estimates.

Exercises

1. Using the teengamb data, model gamble as the response and the other variables as predictors. Take care to investigate the possibility of interactions between sex and the other predictors. Interpret your final model.

2. Using the `infmort` data, find a simple model for the infant mortality in terms of the other variables. Be alert for transformations and unusual points. Interpret your model by explaining what the regression parameter estimates mean.
3. Plot the `ToothGrowth` data with `len` as the response. Fit a linear model to the data and check for possible simplification. Display the fitted regression lines on top of the data.
4. Investigate whether the `side` variable can be used in a model for all the `chredlin` data rather than as a subsetting variable as in the analysis in the text.
5. Find a good model for the `uswages` data with `wages` as the response and all the other variables as predictors.

